

Update the IDE and SDK tools

Once you install Android Studio, you can keep the Android Studio IDE and Android SDK tools up to date with automatic updates and the Android SDK Manager.

Update your IDE using JetBrains Toolbox

If you installed Android Studio using [JetBrains Toolbox](https://www.jetbrains.com/toolbox-app/)

(<https://www.jetbrains.com/toolbox-app/>), then Toolbox is responsible for handling updates to Android Studio. Toolbox lets you install canary, beta, and stable versions of Android Studio in parallel. It also lets you roll back to earlier versions of each, if required. When an update is available it displays in Toolbox, as shown in figure 1.

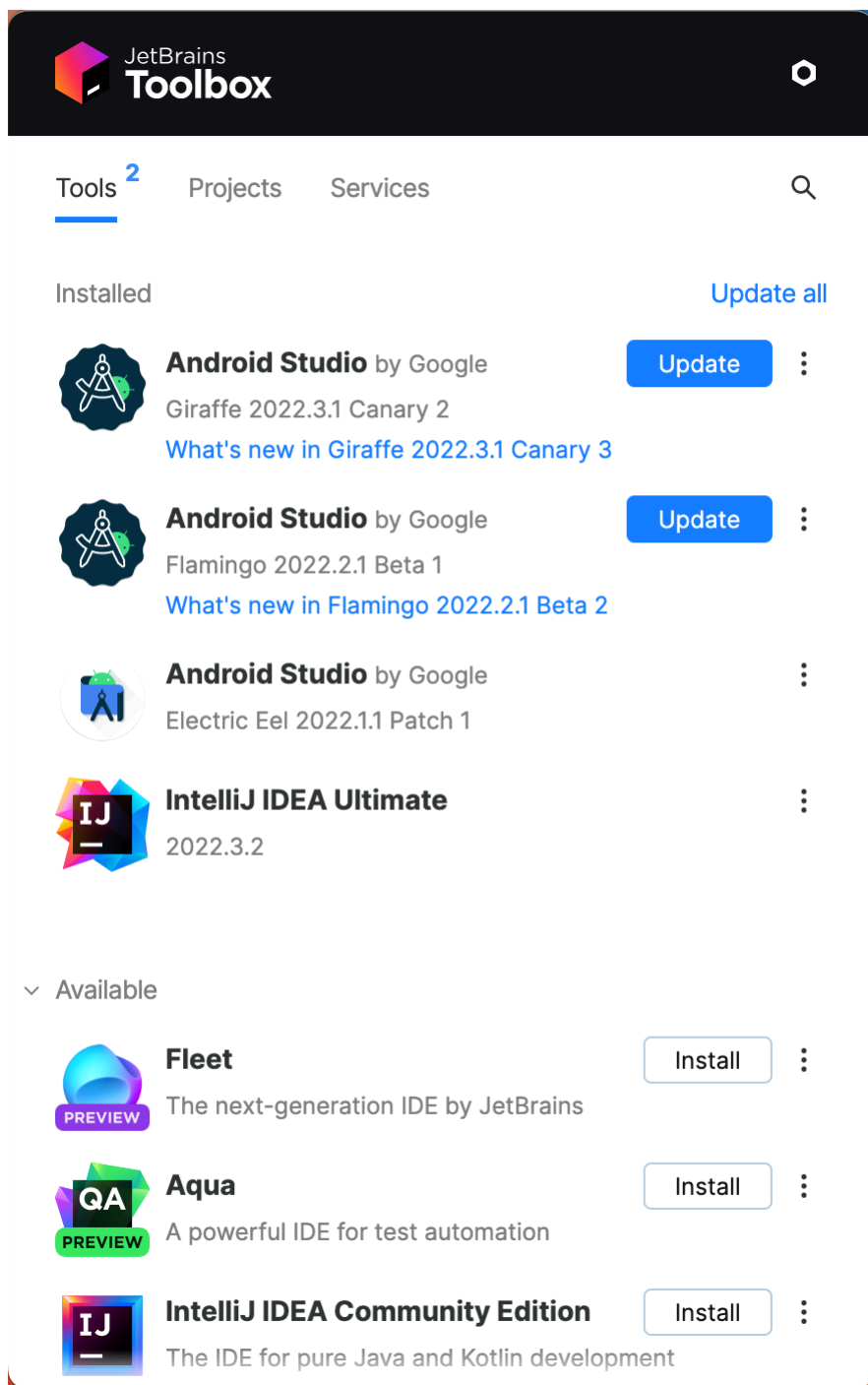


Figure 1. JetBrains Toolbox showing available updates.

Update your IDE and change channels

If you installed Android Studio manually, Android Studio notifies you with a small bubble dialog when an update is available for the IDE. To manually check for updates, click **File** > **Settings** > **Appearance & Behavior** > **System Settings** > **Updates** (on macOS, **Android Studio** > **Check for Updates**). See figure 2.

Updates for Android Studio are available from the following release channels:

- **Canary channel:** these bleeding-edge releases are updated roughly weekly and are available for download on the [Preview release](/studio/preview) (/studio/preview) page.

In addition to receiving canary versions of Android Studio, you also receive preview versions of other SDK tools, including the Android Emulator.

Although these builds are subject to more bugs, they do get tested and are available so you can try new features and provide feedback.



Note: This channel is not recommended for production development.

- **Beta channel:** these are release candidates based on stable canary builds and are available for download on the [Preview release](/studio/preview) (/studio/preview) page. They are released to get feedback before being integrated into the stable channel.
- **Stable channel:** the official, stable release of [Android Studio](/studio) (/studio).

If you'd like to try one of the preview channels (canary or beta) while still using the stable build for your production projects, you can [install them side by side](/studio/preview/install-preview) (/studio/preview/install-preview).

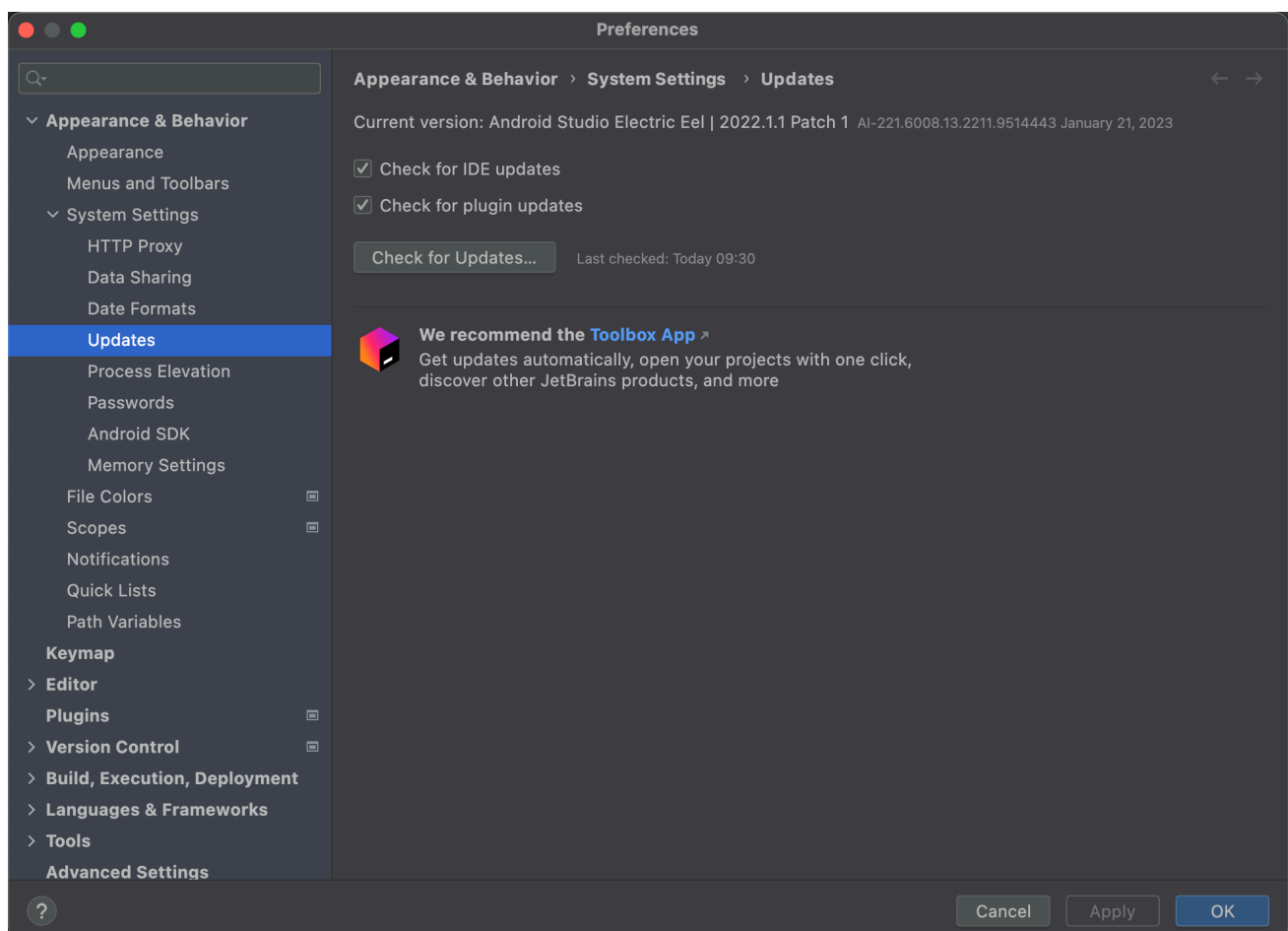
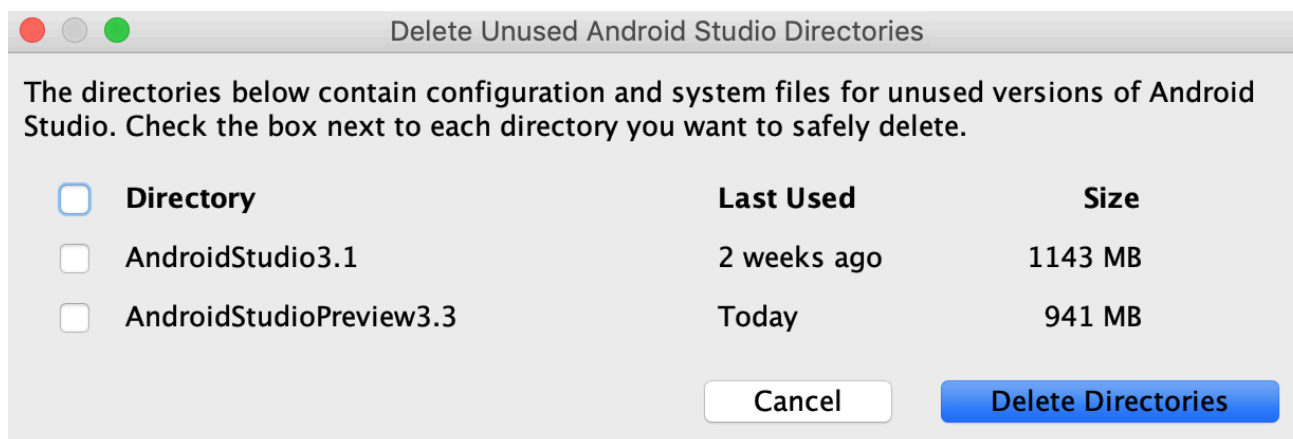


Figure 2. The Android Studio Updates preferences.

Delete unused Android Studio directories



When you run a major version of Android Studio for the first time, it looks for directories containing caches, settings, indices, and logs for versions of Android Studio for which a corresponding installation can't be found. The **Delete Unused Android Studio Directories** dialog then displays locations, sizes, and last-modified times of these unused directories and provides an option to delete them.

Update your tools with the SDK Manager

The Android SDK Manager helps you download the SDK tools, platforms, and other components you need to develop your apps. Once downloaded, you can find each package in the directory indicated as the **Android SDK Location**, as shown in figure 3.

To open the SDK Manager from Android Studio, click **Tools > SDK Manager** or click **SDK Manager**



in the toolbar. If you're not using Android Studio, you can download tools using the [sdkmanager](https://developer.android.com/studio/intro/update#sdk-manager) (/studio/command-line/sdkmanager) command-line tool.

When an update is available for a package you already have, a dash



appears in the checkbox next to the package.

- To update an item or install a new one, select the checkbox.
- To uninstall a package, click to clear the checkbox.

Pending updates are indicated in the left column with a download icon



. Pending removals are indicated with a red X:



To update the selected packages, click **Apply** or **OK** and agree to any license agreements.

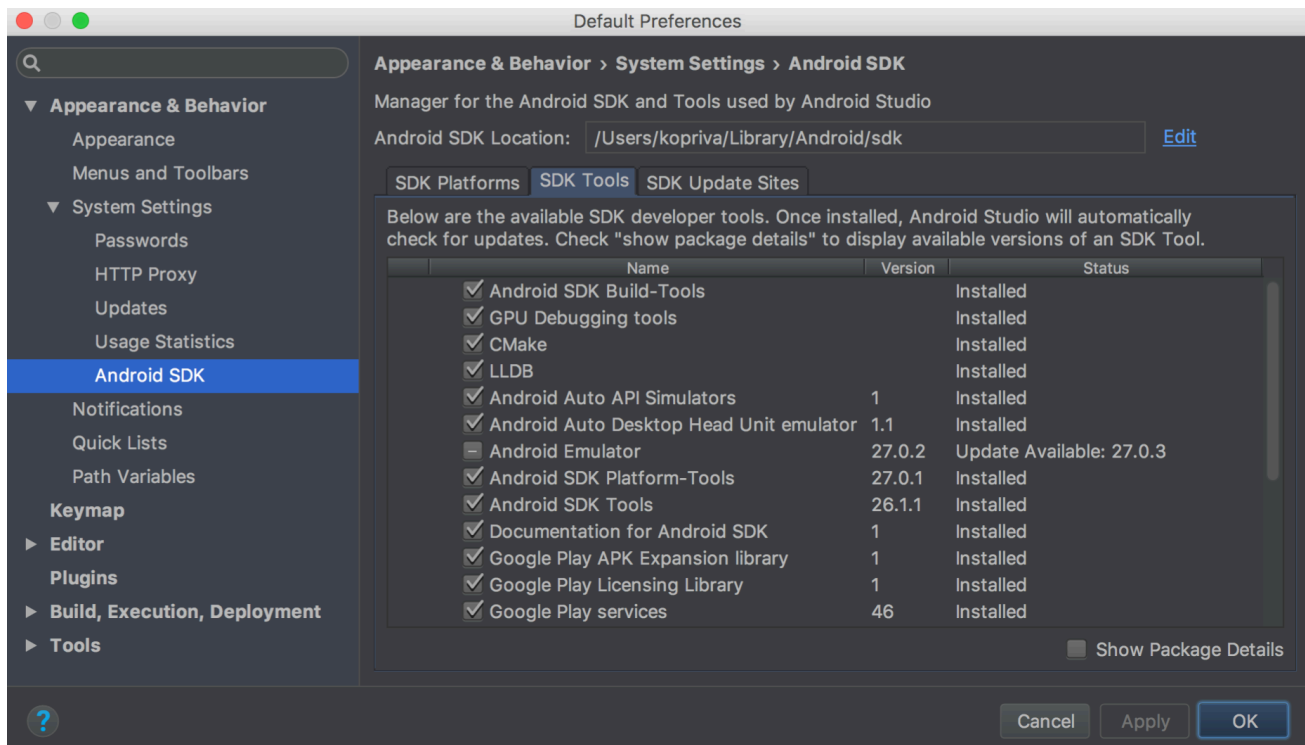


Figure 3. The Android SDK Manager.

Required packages

You can find the following tools in the **SDK Tools** tab:

Android SDK Build Tools

Includes tools to build Android apps. For more information, see the [SDK Build Tools release notes](/studio/releases/build-tools) (/studio/releases/build-tools).

Android SDK Platform Tools

Includes various tools required by the Android platform, including the [adb](/studio/command-line/adb) (/studio/command-line/adb) tool.

Android SDK Command-Line Tools

Includes essential tools such as ProGuard. For more information, see the [SDK Tools release notes](/studio/releases/sdk-tools) (/studio/releases/sdk-tools).

Android SDK Platform

In the **SDK Platforms** tab, you must install at least one version of the Android platform so you can compile your app. Use the latest platform version as your build target to provide the best user experience on the latest devices. To download a version, select the checkbox next to the version name.

You can still run your app on older versions; however, you must build against the latest version to use new features when running on devices with the latest version of Android.

Google USB Driver

Required for Windows. Includes tools to help you perform [adb](#) (/studio/command-line/adb) debugging with Google devices. To install, visit [Get the Google USB Driver](#) (/studio/run/win-usb).

Recommended packages

The following tools are recommended for development:

Android Emulator

A QEMU-based device-emulation tool that you can use to debug and test your applications in an actual Android runtime environment. For more details, see the [Emulator release notes](#) (/studio/releases/emulator).

Note: Most API libraries that were previously provided by the **Support Repository** packages (such as the Android Support Library, Constraint Layout, Google Play services, and Firebase) are now available from Google's Maven repository. Projects created with Android Studio 3.0 and higher automatically include this repository in the build configuration. If you're using an older project, you must manually [add Google's Maven repository](#) (/studio/build/dependencies#google-maven) to your **build.gradle** or **build.gradle.kts** file.

Intel or ARM System Images

The system image is required to run the [Android Emulator](#) (/tools/devices/emulator). Each platform version contains the supported system images. You can also download system images later when creating Android Virtual Devices (AVDs) in the [AVD Manager](#) (/studio/run/managing-avds). Select either Intel or ARM based on your development computer's processor.

Google Play services

Includes a set of libraries, Javadocs, and samples to help build your app. If you want to use APIs from Google Play services (<https://developers.google.com/android/>), you must use either the Google APIs system image or the Google play system image.

The preceding list is not comprehensive, and you can add other sites to download additional packages from third parties, as described in the following section.

In some cases, an SDK package might require a specific minimum revision of another tool. If so, the SDK Manager notifies you with a warning and adds the dependencies to your list of downloads.

Edit or add SDK tool sites

Under the **SDK Update Sites** tab, you can add and manage other sites that host their own tools, and then download the packages from those sites. Android Studio checks for Android tools and third-party tool updates from the SDK sites you add.

For example, a mobile carrier or device manufacturer might offer additional API libraries that are supported by their own Android-powered devices. To develop using their libraries, you can install their Android SDK package by adding their SDK tools URL to the **SDK Manager** in the **SDK Update Sites** tab.

If a carrier or device manufacturer has hosted an SDK add-on repository file on their website, follow these steps to add the site to the Android SDK Manager:

1. Click the **SDK Update Sites** tab.
2. Click **Add**

at the top of the window.
3. Enter the name and URL of the third-party site, then click **OK**.
4. Make sure the checkbox is selected in the **Enabled** column.
5. Click **Apply** or **OK**.

Any SDK packages available from the site now appear in the **SDK Platforms** or **SDK Tools** tabs, as appropriate.

Auto-download missing packages with Gradle

When you run a build from the command line (/studio/build/building-cmdline) or Android Studio, Gradle can automatically download missing SDK packages that a project depends on, as long as the corresponding SDK license agreements have already been accepted in the **SDK Manager**.

When you accept the license agreements using the SDK Manager, Android Studio creates a licenses directory inside the SDK home directory. This licenses directory is necessary for Gradle to auto-download missing packages.

If you have accepted the license agreements on one workstation but want to build your projects on a different one, you can export your licenses by copying over the accepted licenses directory.

To copy the licenses to another machine, follow these steps:

1. On a machine with Android Studio installed, click **Tools > SDK Manager**. At the top of the window, note the **Android SDK Location**.
2. Navigate to that directory and locate the `licenses/` directory inside it.

If you don't see a `licenses/` directory, return to Android Studio, update your SDK tools, and accept the license agreements. When you return to the Android SDK home directory, you should see the directory.

3. Copy the entire `licenses/` directory and paste it into the Android SDK home directory on the machine where you want to build your projects.

Gradle can now automatically download missing packages your project depends on.

Note that this feature is automatically disabled for builds you run from Android Studio, as the SDK manager handles downloading missing packages for the IDE. To manually disable this feature, set `android.builder.sdkDownload=false` in the `gradle.properties` file for your project.

Update your tools with the command line

On systems that don't have a graphical UI, such as CI servers, you can't use the SDK Manager in Android Studio. Instead, use the `sdkmanager` (/studio/command-line/sdkmanager) command-line tool to install (/studio/command-line/sdkmanager#install) and update (/studio/command-line/sdkmanager#update-all) SDK tools and platforms.

After installing SDK tools and platforms using `sdkmanager`, you might need to accept any missing licenses. This can also be done using `sdkmanager`:


```
$ sdkmanager --licenses
```

This command scans all of the installed SDK tools and platforms and displays any licenses that have not been accepted. You are prompted to accept each license.

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